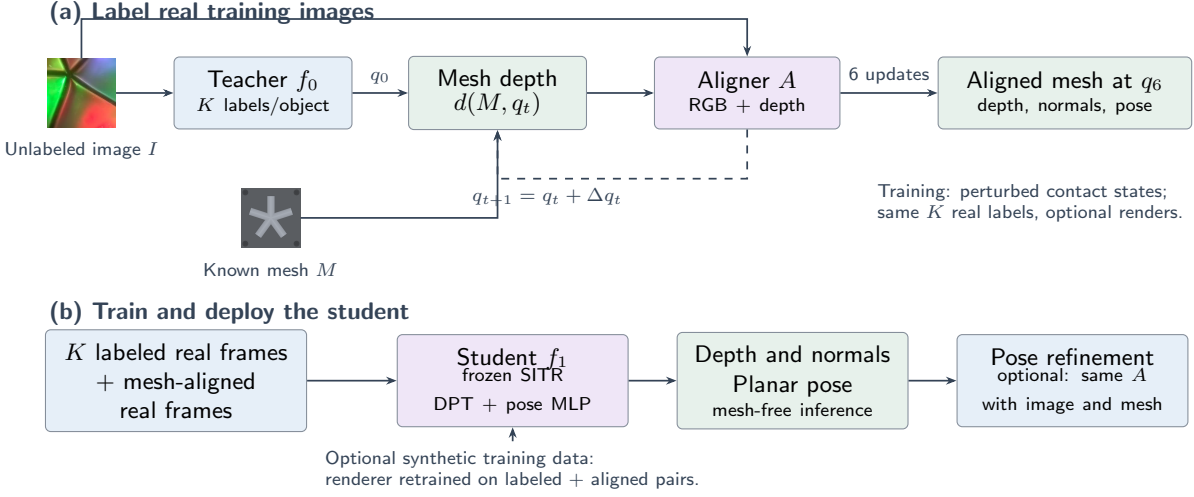

(a) Label real training images
Unlabeled image I
Teacher f_0
K labels/object
q_0
Mesh depth
$d(M, q_t)$
Aligner A
RGB + depth
6 updates
Aligned mesh at q_6
depth, normals, pose
$q_{t+1} = q_t + \Delta q_t$
Known mesh M
Training: perturbed contact states;
same K real labels, optional renders.
(b) Train and deploy the student
K labeled real frames
+ mesh-aligned
real frames
Student f_1
frozen SITR
DPT + pose MLP
Depth and normals
Planar pose
mesh-free inference
Pose refinement
optional: same A
with image and mesh
Optional synthetic training data:
renderer retrained on labeled + aligned pairs.